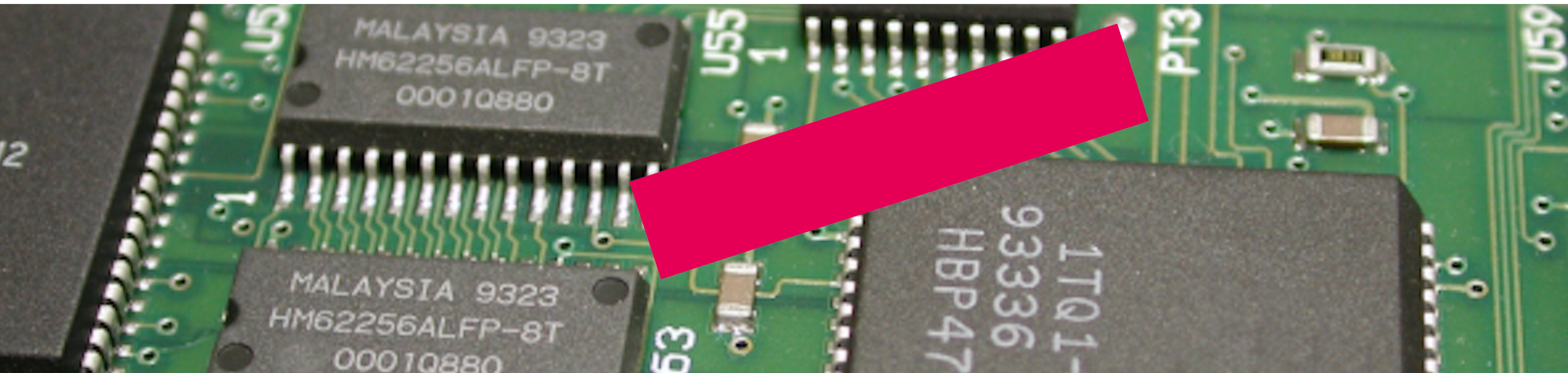


Prog 5 Embedded Systems Development - 5. Concurrency and more



Embedded Systems Engineering
School of Engineering and Automotive

Johan.Korten@han.nl

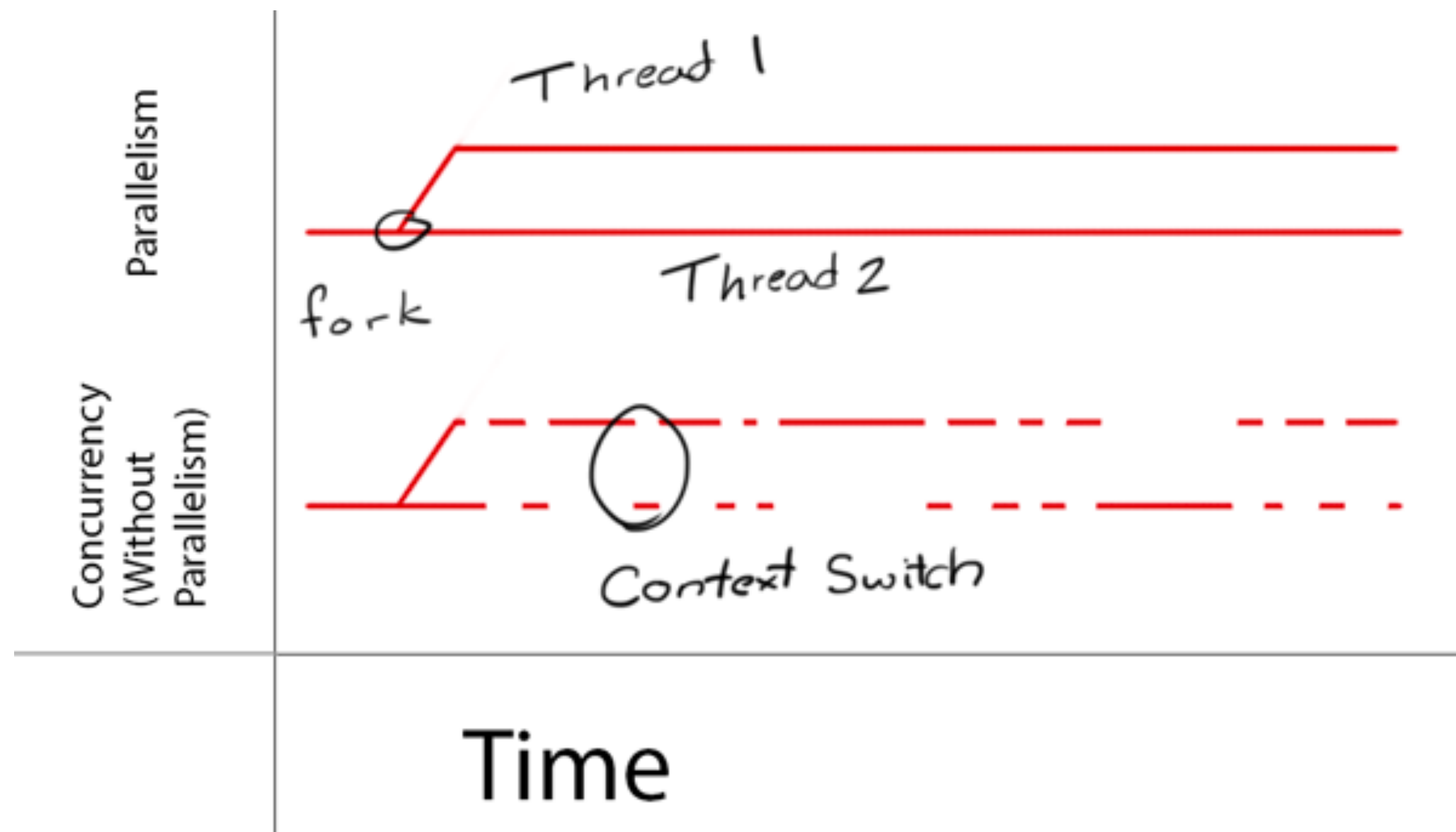
To begin with...

Attendance

Schedule (exact info see #00 and roster at insite.han.nl)

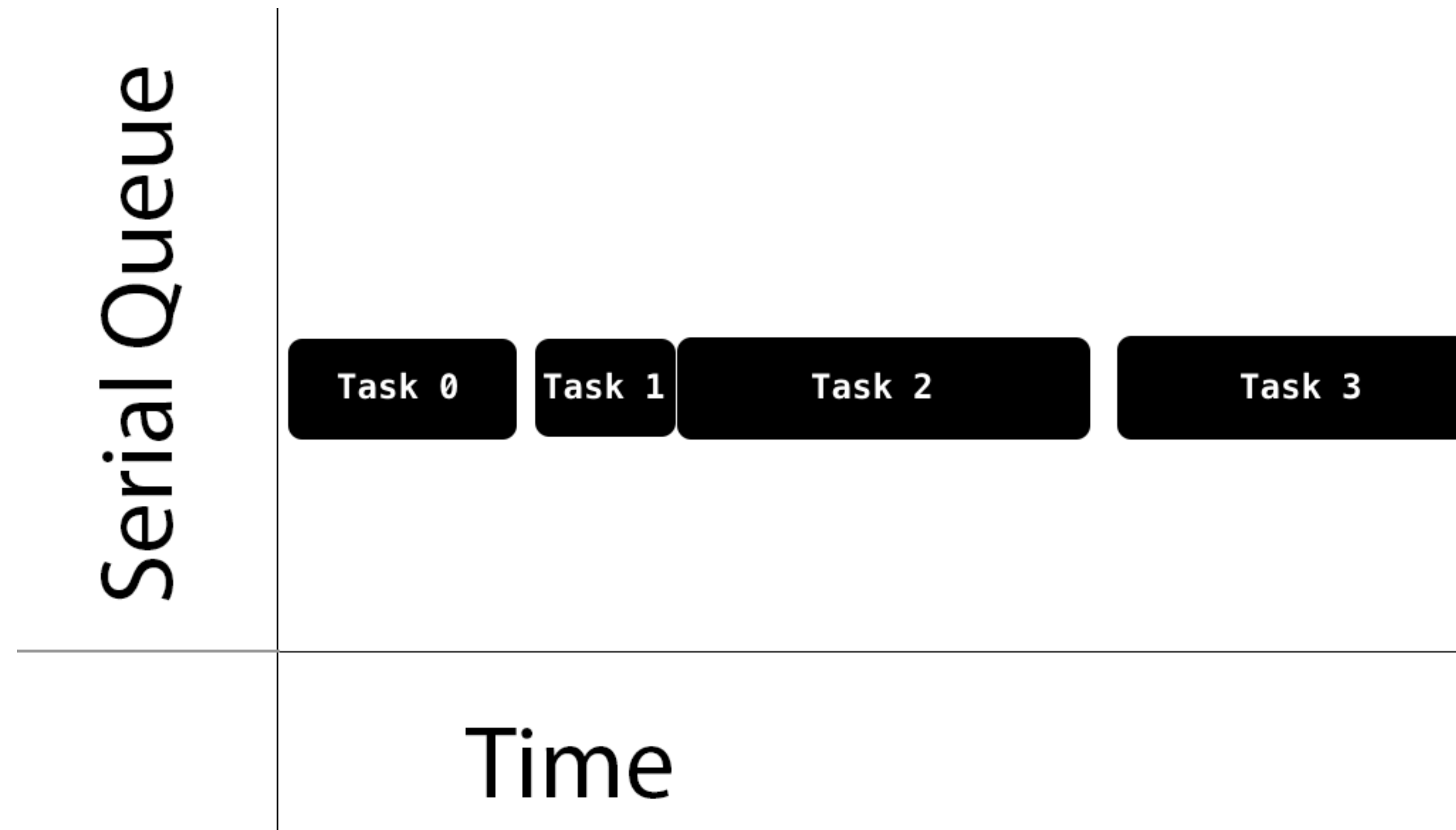
Week (Thu)	Design principle	C++	UML	Lab (Fri)
Step 1 - 3 Sep	Single Responsibility	Scope, namespaces, string	Use cases / class diagram	4 Sep: Git
Step 2 - 10 Sep	Open-Closed Principle	Constructors, iterators, lambdas	Inheritance / generalization	11 Sep: Requirements
Step 3 - 17 Sep	Liskov Substitution Principle	Lists, inline functions, default params	Activity diagrams	18 Sep: sensor library 1: core, Bus, Arduino
Step 4 - 24 Sep	Interface Segregation Principle	Interfaces and abstract classes	Dependencies	25 Sep: sensor library 2: Raspberry Pi, Bus / Clock split
Step 5 - 1 Oct	Dependency Inversion Principle	Threads, callbacks	Sequence diagrams	2 Oct: sensor library 3: sampler thread, callback, MQTT
Step 6 - 8 Oct	Coupling and cohesion	Polymorphism	Composition, packages	9 Oct: sensor library 4: polymorphism, UML design
Step 7 - 15 Oct	Embedded SOLID	Lifecycle: constructors, destructors, const	-	16 Oct: sensor library 5: RAI, host unit tests, refactor

Concurrency



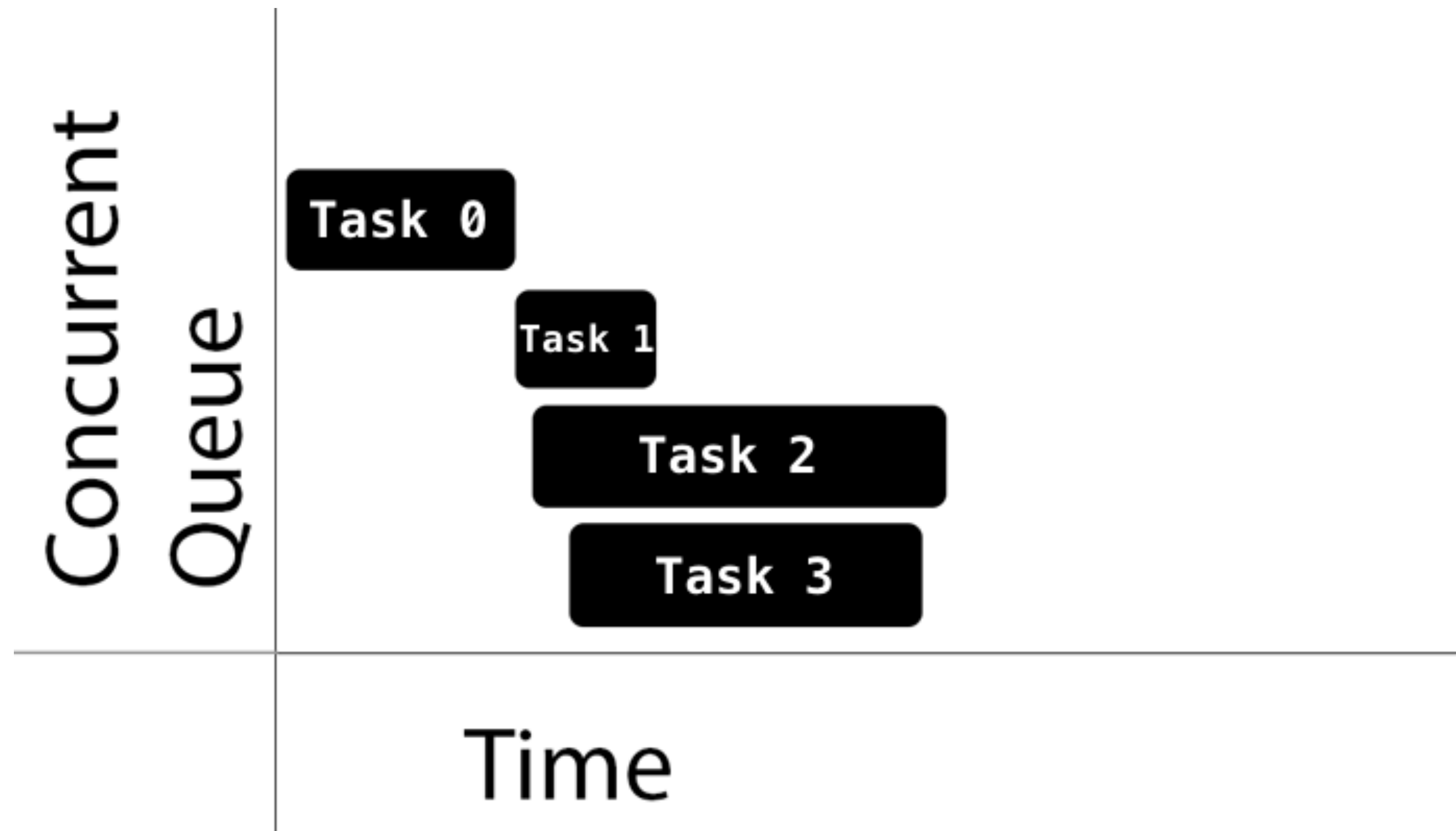
Source: Ray Wenderlich

Queues: Serial Queue



Source: Ray Wenderlich

Queues: Concurrent Queue



Source: Ray Wenderlich

Who calls the sensor, and when?

```
// week 4 smoke test
while (keepRunning) {
    sensor.readForced(m);
    print(m);
    std::this_thread::sleep_for(std::chrono::seconds(1));    // main is stuck here
}
```

```
// Problems:
// - main can do nothing else (no MQTT keep-alive, no command input)
// - the timing is mixed with the printing
// - nobody else can be told about a new measurement
```

Sampler: a thread that owns the timing

```
class Sampler {
public:
    using Callback = std::function<void(const Measurement&)>;

    Sampler(EnvironmentSensor& sensor, std::chrono::milliseconds period);
    ~Sampler();                // stop()

    void onMeasurement(Callback cb);    // runs on the sampler thread!
    void start();
    void stop();                    // joins, wakes the sleep early

private:
    void run();                  // loop: readForced -> callback -> wait
    EnvironmentSensor&          sensor_;
    Callback                   callback_;
    std::thread                 thread_;
    std::atomic<bool>           running_{false};
    std::condition_variable     wakeUp_;
};
```


C++ Language: Threads (C++11 and higher)

```
#include <thread>
```

C++ Language: Threads (C++11 and higher): Callable function

```
#include <thread>

// function to be used in callable

void callable_func(int N)
{
    for (int i = 0; i < N; i++) {
        cout << "Thread 1 :: callable => function pointer\n";
    }
}
```

C++ Language: Threads (C++11 and higher): Callable Object

```
#include <thread>

// A callable object
class thread_obj {
public:
    void operator()(int n) {
        for (int i = 0; i < n; i++){
            cout << "Thread 2 :: callable => function object\n";
        }
    }
};
```


C++ Language: Threads (C++11 and higher): Lambda Expression

```
#include <thread>

// Define a Lambda Expression
auto f = [](int n) {
    for (int i = 0; i < n; i++) {
        cout << "Thread 3 :: callable => lambda expression\n";
    }
};
```


C++ Language: Threads (C++11 and higher): Lambda Expression

```
#include <thread>

int main()
{
    // ... e.g. Lambda expression

    // launch thread using function pointer as callable
    thread th1(collable_func, 2);

    // launch thread using function object as callable
    thread th2(thread_obj(), 2);

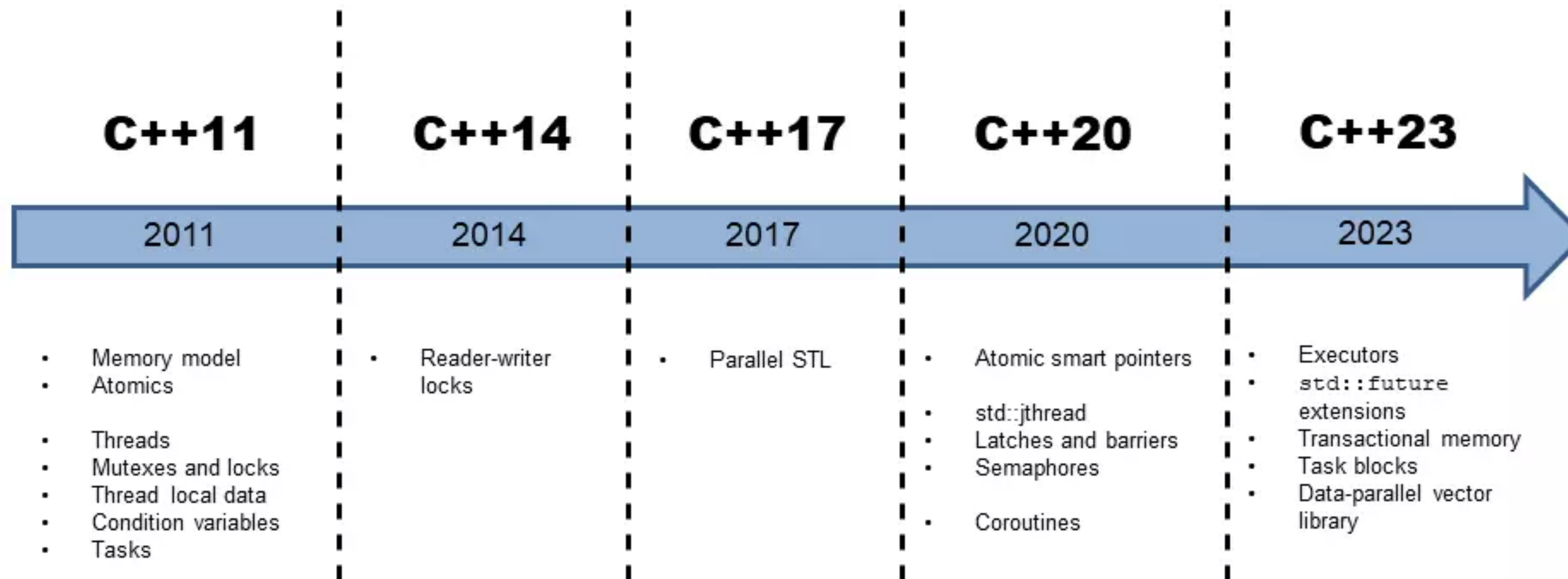
    // launch thread using lambda expression as callable
    thread th3(f, 2);

    // Wait for thread t1 to finish
    th1.join();
    // Wait for thread t2 to finish
    th2.join();

    // Wait for thread t3 to finish
    th3.join();

    return 0;
}
```

C++ Language: 'Concurrency Evolution'



Typical Concurrency Problems / Solutions

Problems:

- Resource contention (dining philosophers) / Race Conditions (e.g. data race)
- Starvation and Deadlock (ultimate form)
- Flight booking system (locking)

<https://youtu.be/NbwbQQB7xNQ>

Solutions:

- Atomic operations
- Locks
- Mutual Exclusives (e.g. semaphores)
- Priority queues

C++ Language Pattern: Singleton

A singleton pattern restricts the instantiation of a class to one "single" instance.

You will be tempted to make the MQTT client a singleton. Do not: it hides the dependency, makes the class untestable and is a global in disguise. Pass a Publisher& instead (Dependency Inversion, later this lecture).

Singleton
- <u>_ singleton : Singleton</u>
- Singleton()
+ <u>getInstance() : Singleton</u>

C++ Language: Callback methods Blocking

```
#include <iostream>
#include <functional>

void processData(int data, std::function<void(int)> callback) {
    // Do some processing on the data
    int processedData = data * 2;

    // Invoke the callback function with the processed data
    callback(processedData);
}

int main() {
    // Define a lambda function to be used as a callback
    auto printResult = [](int result) {
        std::cout << "Processed data: " << result << std::endl;
    };

    // Pass the lambda as a callback to the processData function
    processData(10, printResult);

    return 0;
}
```

Tip: use breakpoints to check the actual flow!

C++ Language: Callback methods Non-Blocking

```
1  #include <iostream>
2  #include <functional>
3  #include <future> // For async
4
5  std::future<void> processDataAsync(int data, std::function<void(int)> callback) {
6      // Use std::async to launch asynchronous task and return a future
7      return std::async( policy: std::launch::async, f: [data, callback]() -> void {
8          // Simulate some asynchronous processing with a delay
9          std::this_thread::sleep_for( d: std::chrono::seconds( r: 10));
10
11          // Process the data
12          int processedData = data * 2;
13
14          // Call the callback function with processed data
15          callback(processedData);
16      });
17 }
```

Tip: use breakpoints to check the actual flow!

C++ Language: Callback methods Non-Blocking

```
18
19 ▶ int main() {
20     // Define a lambda function to be used as a callback
21     auto printResult : void (int) const = [](int result) -> void {
22         std::cout << "Processed data: " << result << std::endl;
23     };
24
25     // Call the asynchronous version of processData and get the future
26     std::cout << "Starting asynchronous processing..." << std::endl;
27     std::future<void> asyncTask = processDataAsync( data: 20, callback: printResult);
28
29     // Do other things while waiting
30     std::cout << "Main thread is free to do other work..." << std::endl;
31
32     // Wait for the async task to complete (if necessary)
33     asyncTask.wait(); // This waits for the asynchronous task to finish
34
35     std::cout << "Asynchronous task finished." << std::endl;
36     return 0;
37 }
38 |
```

Tip: use breakpoints to check the actual flow!

C++ Language: Callback methods Non-Blocking

```
18
19 ► int main() {
20     // Define a lambda function to be used as a callback
21     auto printResult : void (int) const = [](int result) -> void {
22         std::cout << "Processed data: " << result << std::endl;
23     };
24
25     // Call the asynchronous version of processData and get the future
26     /Users/jakorten/Development/CallbacksCPP_Nonblock/cmake-build-debug/CallbacksCPP_Nonblock
27     Starting asynchronous processing...
28     Main thread is free to do other work...
29     Processed data: 40
30     Asynchronous task finished.
31
32     Process finished with exit code 0
33 ● asyncTask.wait(); // This waits for the asynchronous task to finish
34
35     std::cout << "Asynchronous task finished." << std::endl;
36     return 0;
37 }
38 |
```

Output of our code...

Our code will work in C++11, C++14, C++17, C++20, and even C++23 without any modifications because it relies on core features (std::async, std::future, and lambdas) that were introduced in C++11 and have been stable across subsequent standards.

C++ Language: Async Callback

std::async() does following things:

- It automatically creates a thread (Or picks from internal thread pool) and a promise object for us.*
- Then passes the std::promise object to thread function and returns the associated std::future object.*
- When our passed argument function exits then its value will be set in this promise object, so eventually return value will be available in std::future object.*

See *async_callback.cpp* and *async_single.cpp* examples.

Callbacks: three forms, one use

```
// 1. free function
void printIt(const Measurement& m) { ... }
sampler.onMeasurement(printIt);

// 2. function object
struct Logger { void operator()(const Measurement& m) { ... } };
sampler.onMeasurement(Logger{});

// 3. lambda: captures what it needs, no globals
sampler.onMeasurement( [&](const Measurement& m) {
    publisher.publish(topic, toJson(m));
});

// std::function<void(const Measurement&)> accepts all three.
```

Threads: what can go wrong here?

The callback runs on the sampler thread, not on main.

Walk through station_main.cpp:

- which objects does the sampler thread touch? sensor, publisher
- which does main touch after start()? none, on purpose

What if main also called sensor.readForced()?

- data race on the Bosch struct and on the I2C transaction
- the sensor class is not thread-safe and should not be
- ownership solves it, not a mutex: one owner, one thread

MqttPublisher::publish() may be called from the sampler thread: mosquitto guards its own queue.
ConsolePublisher: one printf, fine.

Revisiting SOLID

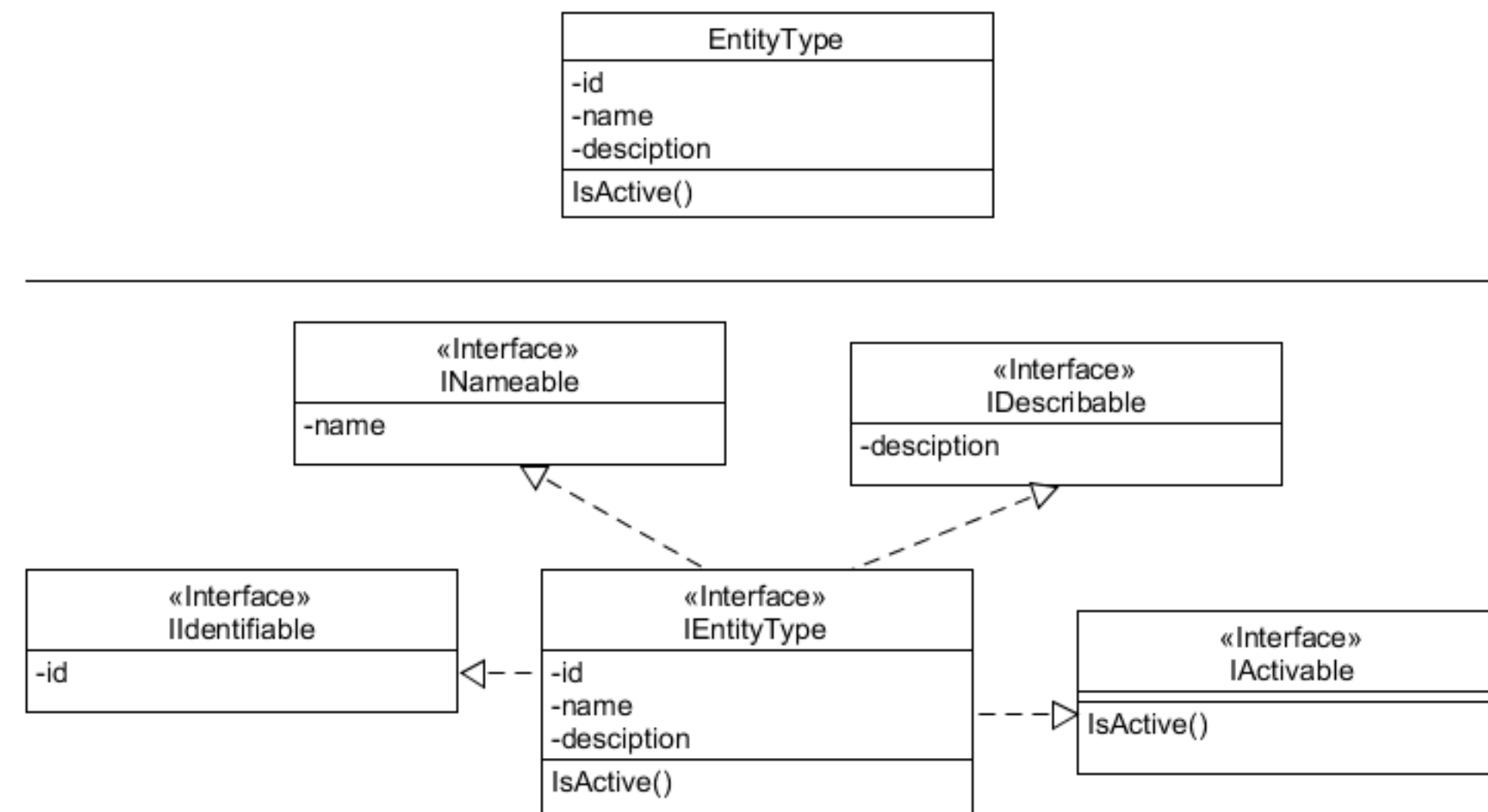
SOLID: Interface Segregation

“No client should be forced to depend on methods it does not use.”

- Uncle Bob

<https://claudiorivera.net/2018/02/04/isp-interface-segregation-principle/>

SOLID: Interface Segregation



<https://claudiorivera.net/2018/02/04/isp-interface-segregation-principle/>

SOLID: Dependency Inversion Principle

“High-level modules should not depend on low-level modules.

Both should depend on abstractions.”

“Abstractions should not depend on details.

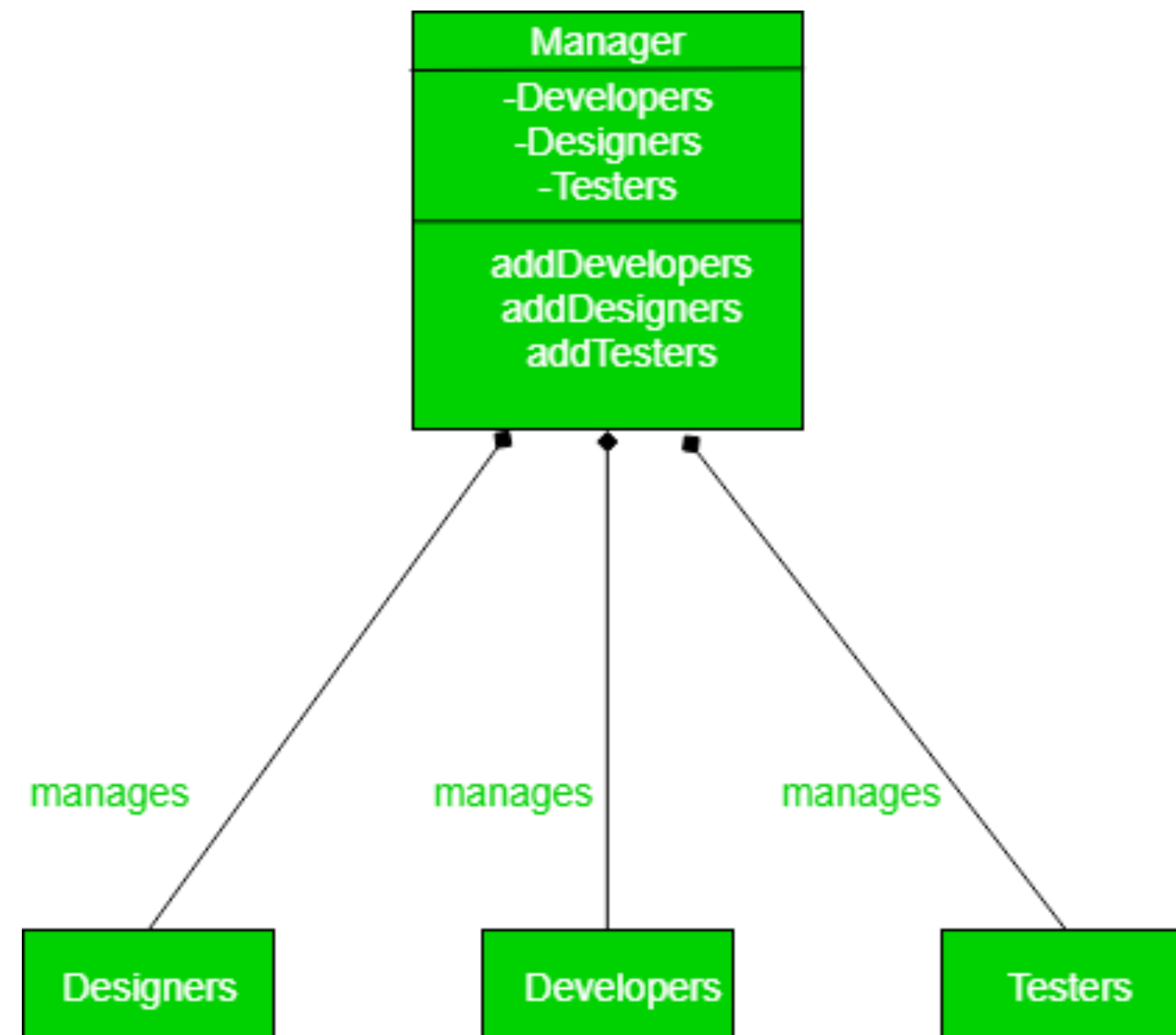
Details should depend on abstractions.”

—R. Martin

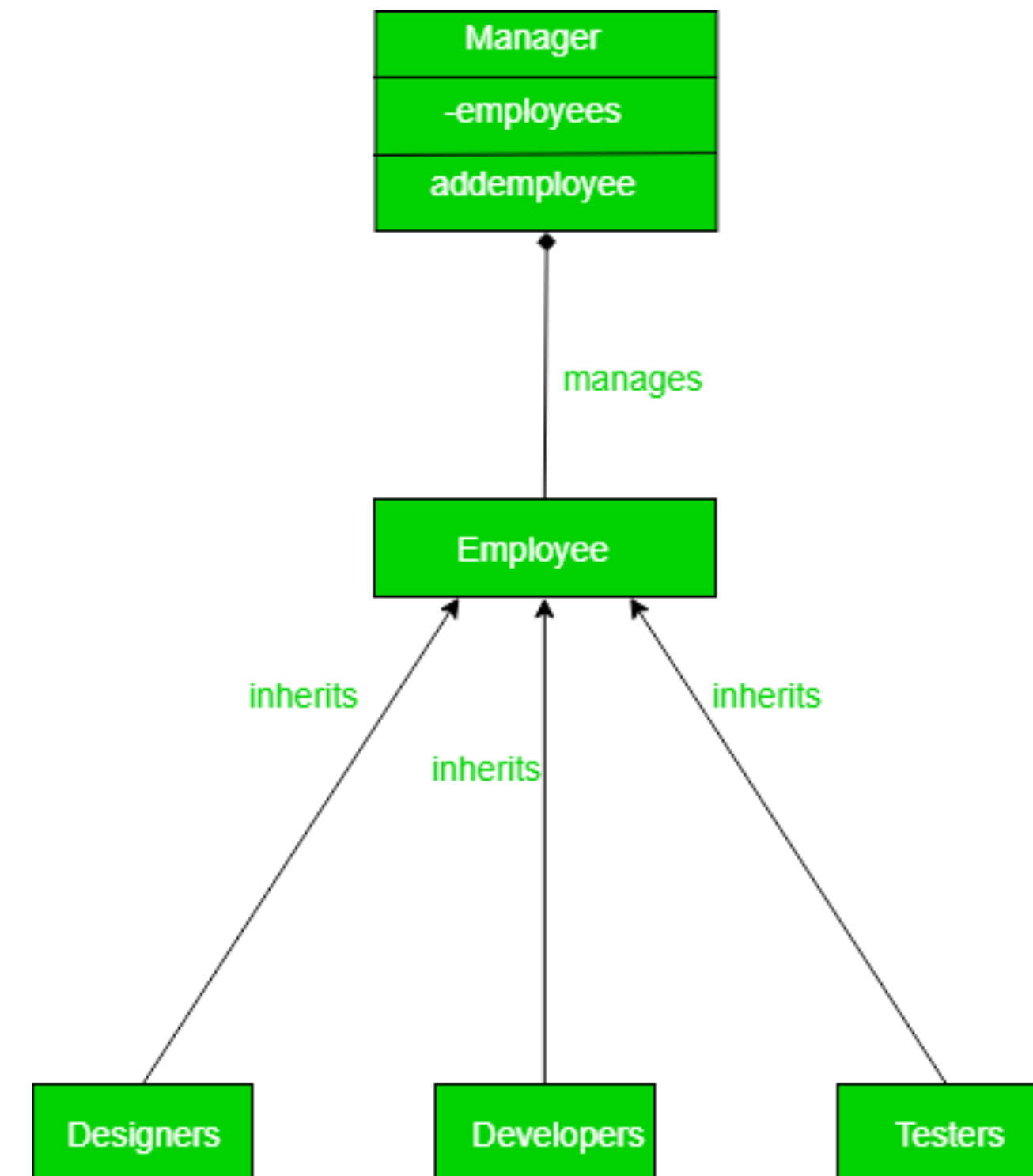
<https://claudiorivera.net/2018/02/12/dip-dependency-inversion-principle/>

SOLID: Dependency Inversion Principle

From:



To:



<https://www.geeksforgeeks.org/dependency-inversion-principle-solid/>

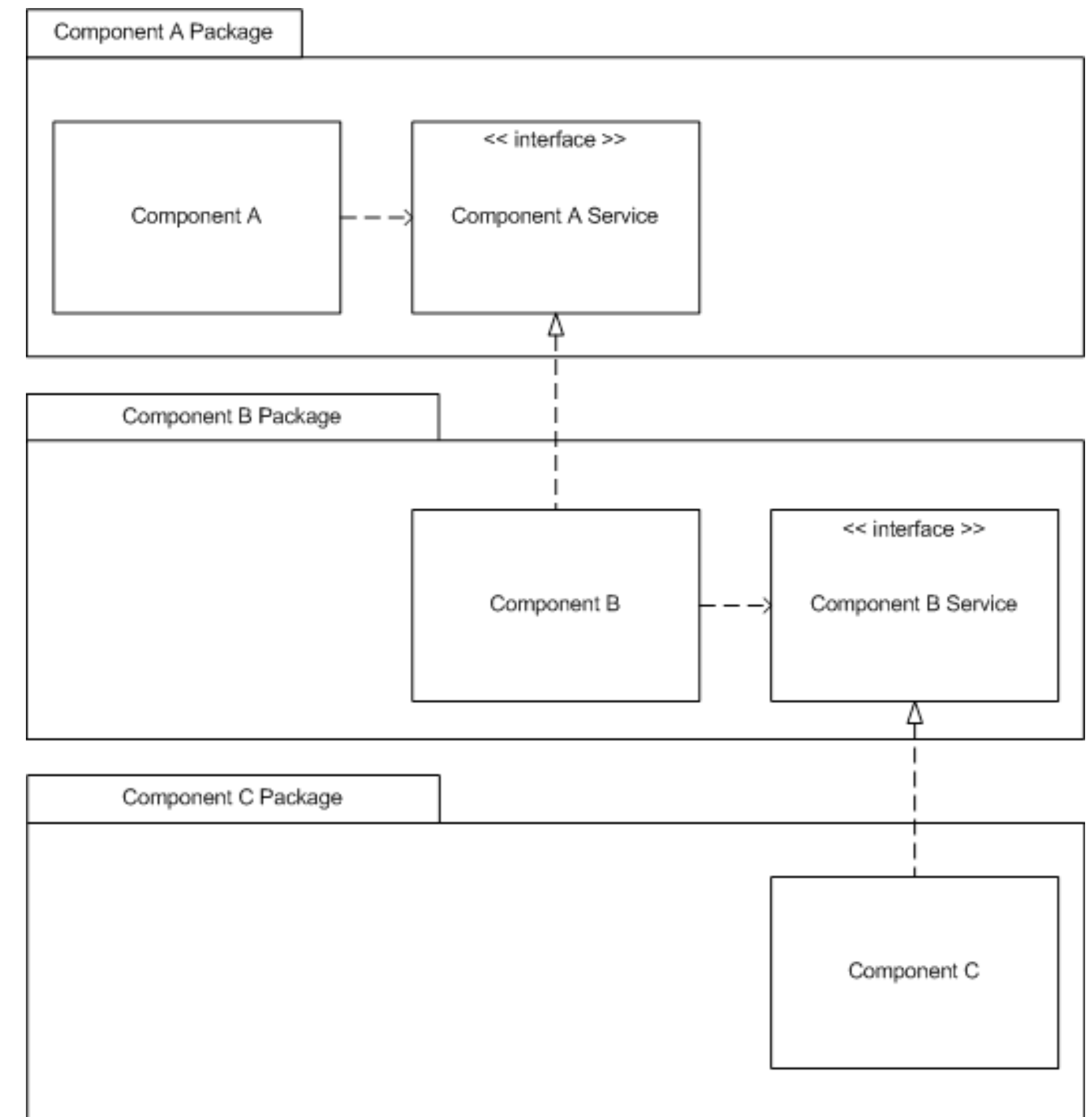
SOLID: Dependency Inversion Principle

Helps to sustain low-coupling between the components comprising an application.

A. High-level modules should not depend upon low-level modules. Both should depend upon abstractions.

B. Abstractions should not depend upon details. Details should depend upon abstractions.

<https://dzone.com/articles/the-dependency-inversion-principle-for-beginners>



Dependency Inversion, applied twice

```
// before: high-level depends on low-level
class Sampler { Sampler(bmp280::Bmp280& sensor, ...); };

// after: both depend on an abstraction
class EnvironmentSensor {
public:
    virtual Error init() = 0;
    virtual Error readForced(Measurement& out) = 0;
};
class Sampler { Sampler(EnvironmentSensor& sensor, ...); };

// and again for the output side
class Publisher {
public:
    virtual bool publish(const std::string& topic, const std::string& payload) = 0;
};
// MqttPublisher, ConsolePublisher, RecordingPublisher (tests)
```


FakeSensor: the whole chain on a laptop

```
class FakeSensor final : public EnvironmentSensor {
    Error init() override { return Error::None; }
    Error readForced(Measurement& out) override {
        out.temperatureC = 20.0F + (tick_++ % 50) * 0.1F;    // a ramp
        out.pressurePa    = 101325.0F;
        return Error::None;
    }
};
```

```
$ ./station_main --fake --console
han/ese/student/bme280/state {"t":20.00,"p":101325}
han/ese/student/bme280/state {"t":20.10,"p":101325}
```

```
// No Pi, no sensor, no broker. That is what the interface buys you.
```

Composition root: the one place where concrete classes meet

```
int main() {
    rpi::LinuxI2cBus bus("/dev/i2c-1", 0x76);
    rpi::LinuxClock clock;
    bme280::Bme280 sensor(bus, clock);
    MqttPublisher publisher("localhost", 1883);

    Sampler sampler(sensor, std::chrono::seconds(1));
    sampler.onMeasurement([&](const Measurement& m) {
        publisher.publish("han/ese/<name>/bme280/state", toJson(m));
    });

    sampler.start();
    waitForCtrlC();
    sampler.stop();
} // destructors: thread joined, MQTT disconnected, /dev/i2c-1 closed

// No #ifdef in the driver. No globals. No singleton.
```

SOLID: Liskov Substitution Principle

Takeaway Points:

- Subtypes must be substitutable for their base types without altering the correctness of the program.
- Derived classes should extend the behavior of base classes without changing their originally intended function.
- Violating this principle can lead to problems with class hierarchies and inheritance, where derived classes are not true extensions of their base classes.

SOLID: Interface Segregation Principle

Takeaway Points:

- Clients should not be forced to depend on interfaces they do not use.
- It's better to have several specific interfaces rather than a one-size-fits-all "general purpose" interface.
- By segregating interfaces, you can ensure that a class only needs to know about the methods that are of interest to it.

SOLID: Dependency Inversion Principle

Takeaway Points:

- High-level modules should not depend on low-level modules. Both should depend on abstractions.
- Abstractions should not depend on details. Details should depend on abstractions.
- By depending on abstractions rather than concrete implementations, you can create more modular and adaptable code.

SOLID: Recap

S — Single Responsibility

Each class has one reason to change.

Avoid: “God” classes mixing hardware, business logic, and persistence.

O — Open/Closed

Open for extension, closed for modification.

Avoid: Ripple edits across the codebase to add a new behavior.

L — Liskov Substitution

Subtypes must be substitutable for their base types.

Avoid: Hardware implementations that break tests when swapped in/out.

I — Interface Segregation

Prefer small, client-specific interfaces.

Avoid: Fat interfaces that mix unrelated concerns.

D — Dependency Inversion

Depend on abstractions, not concretions.

Avoid: High-level policy tied directly to low-level hardware details.

SOLID → Loose Coupling, High Cohesion

SOLID: Main reasons

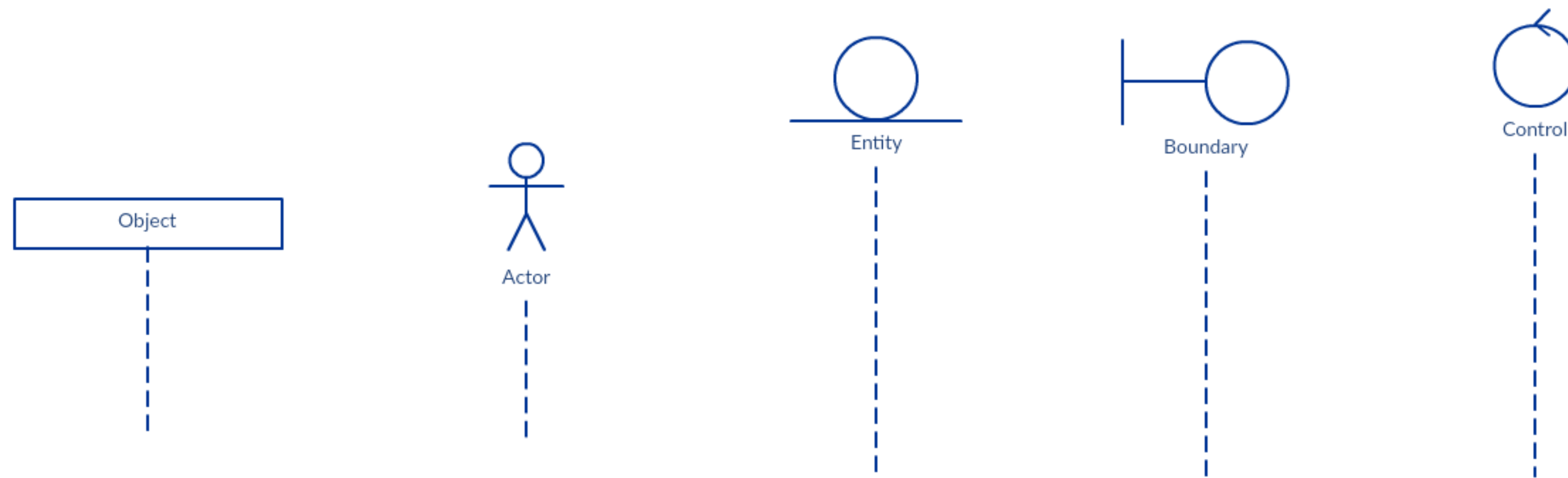
Takeaway Points:

- Following the SOLID principles:
- can greatly improve the quality of your object-oriented design and architecture.
- makes your code more robust, maintainable, and extensible
- leads to software that is easier to understand, modify, and enhance.

So: Learn to use it as much as possible!

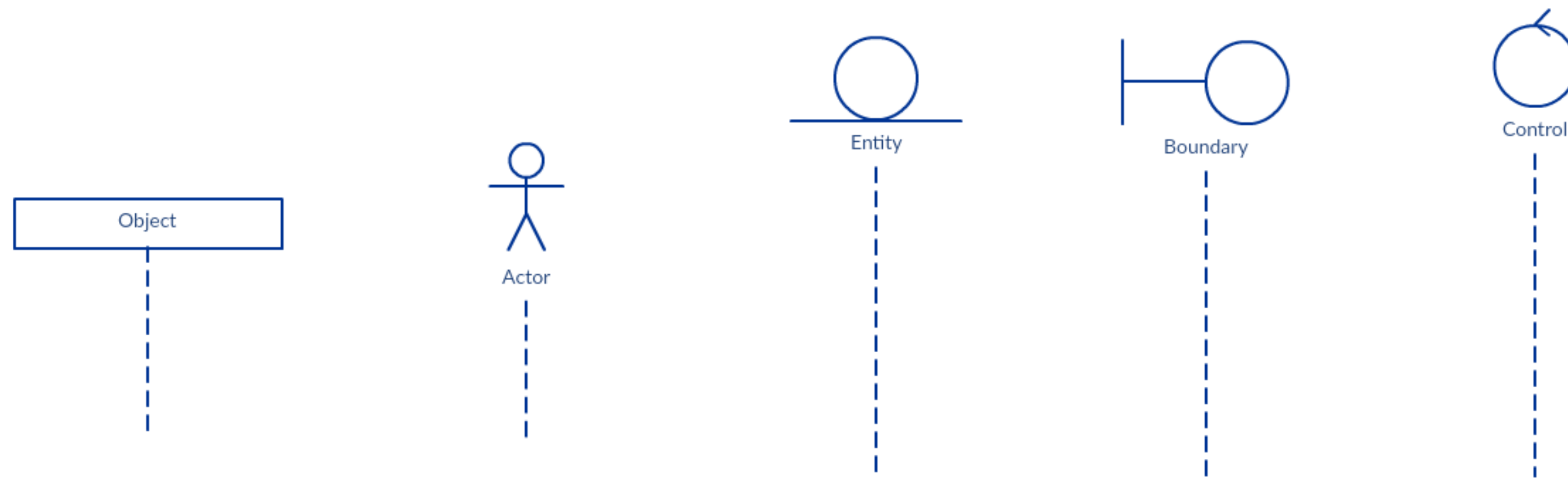
We will talk later about: applying solid in an Embedded Context

UML: Sequence Diagram - Stereotypes



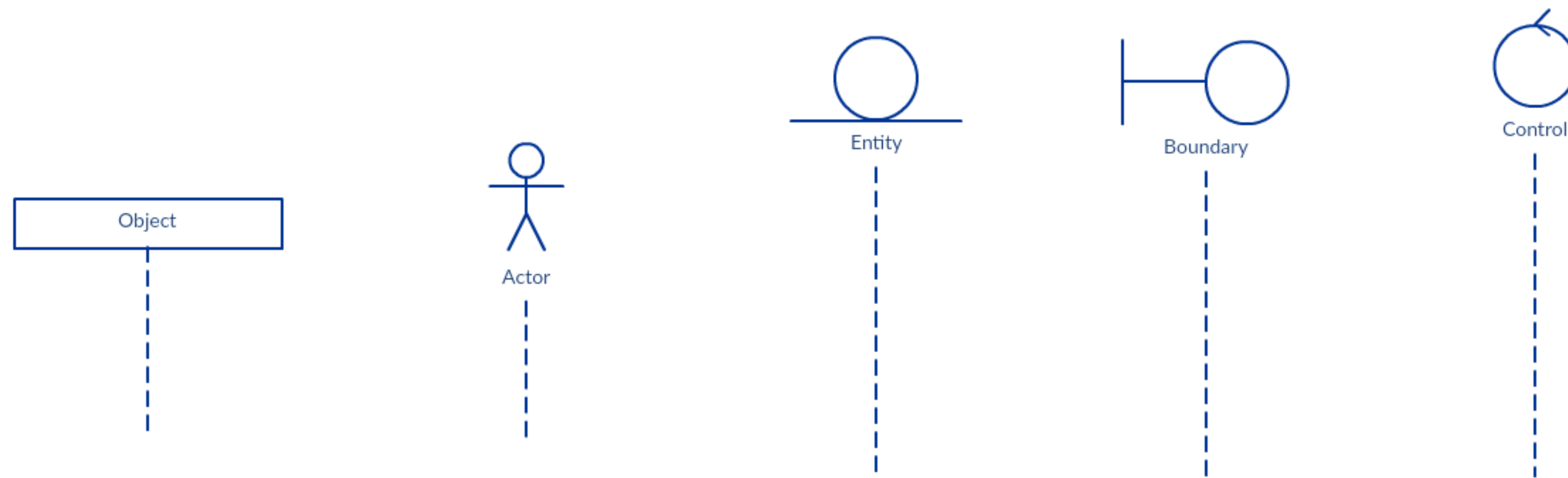
Typical **stereotypes** used in a Sequence Diagram.

UML: Sequence Diagram - Object



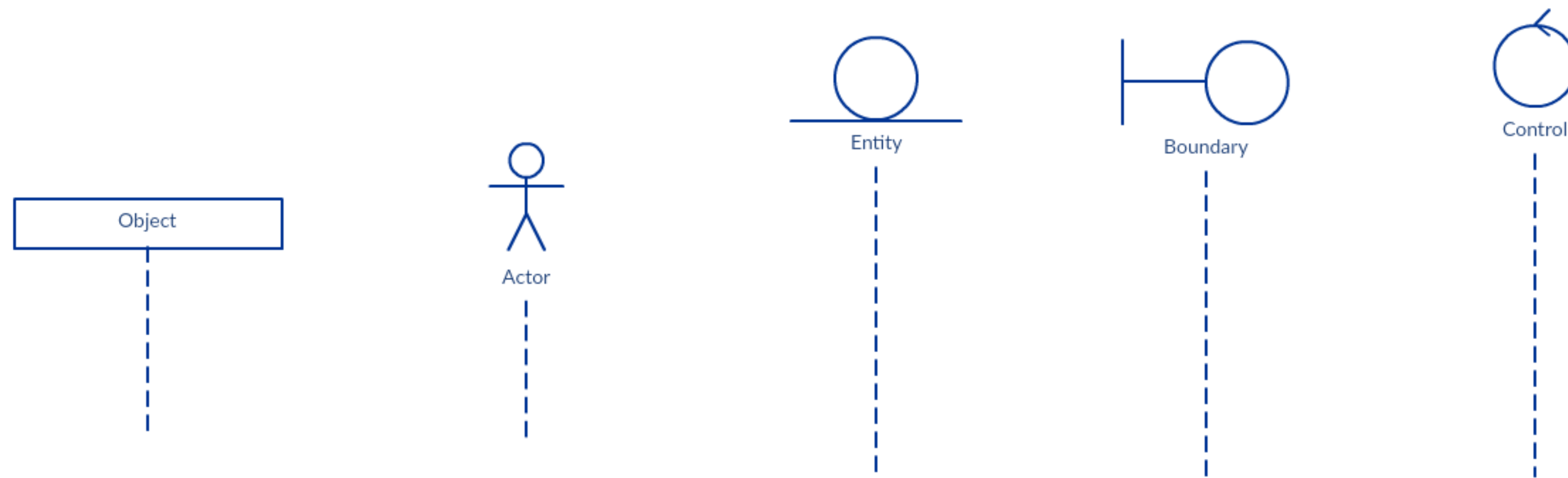
An **object** represents an *instance* of a class.

UML: Sequence Diagram - Actor



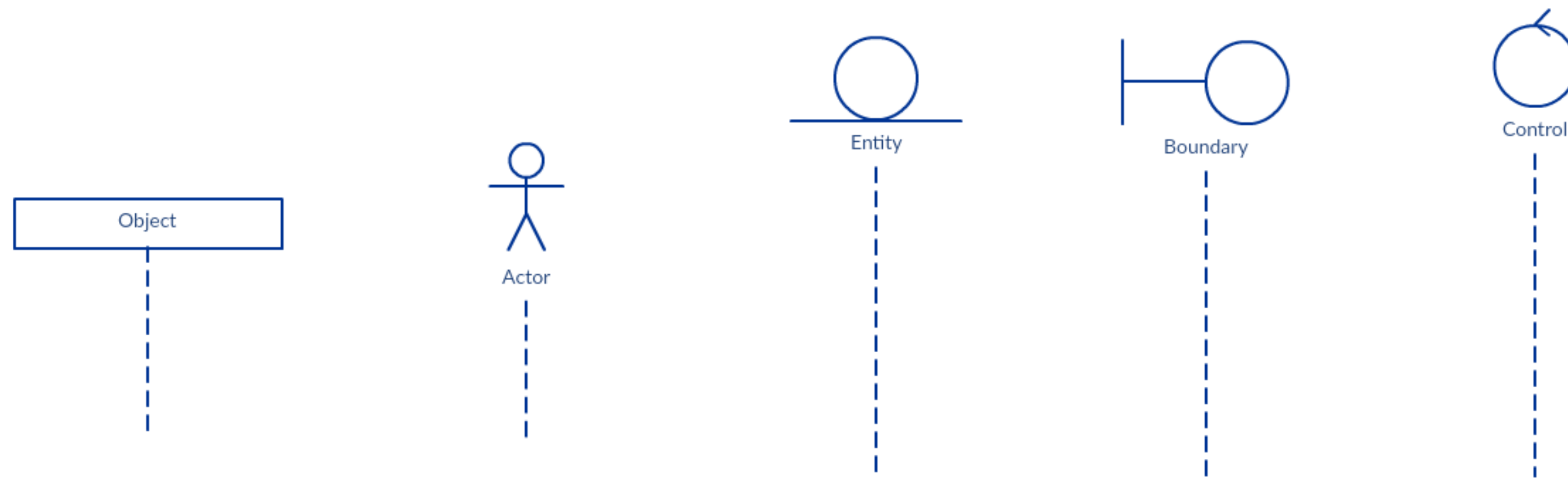
An **actor** represents a *user* in the system or an *other system* interacting with the current system

UML: Sequence Diagram - Entity



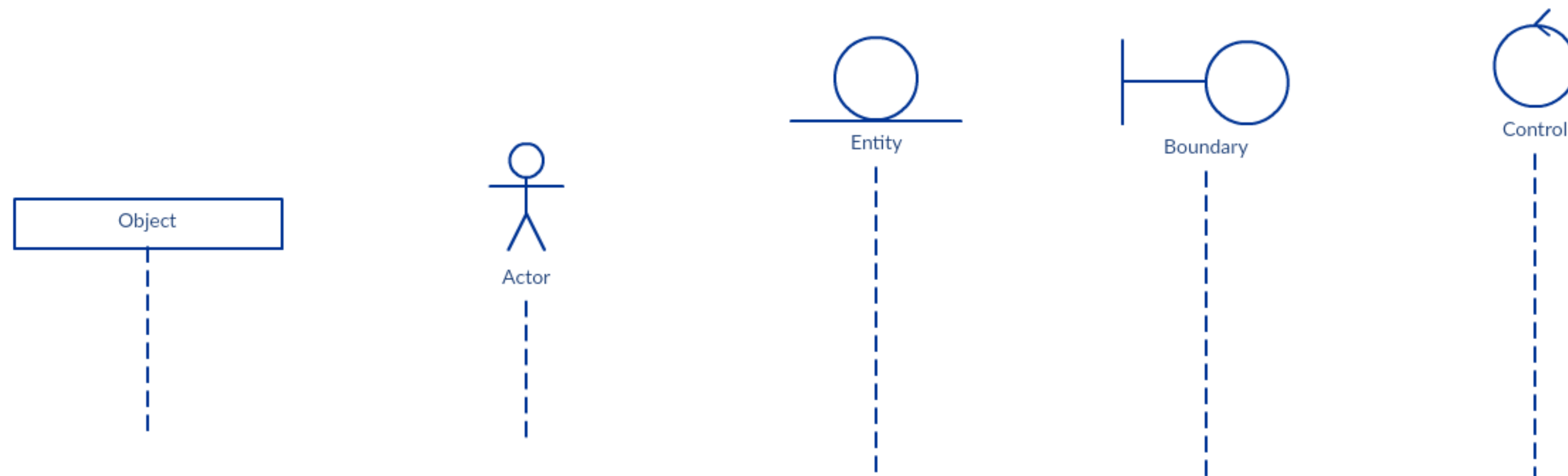
An **entity** is any singular, identifiable and separate object. It refers to individuals, organizations, systems, bits of data or even distinct system components that are considered significant in and of themselves.

UML: Sequence Diagram - Boundary



A **boundary** is a stereotyped Object that models some *system boundary*, typically a user interface screen. In an MVC (Model-View-Controller) pattern it represents the *View*.

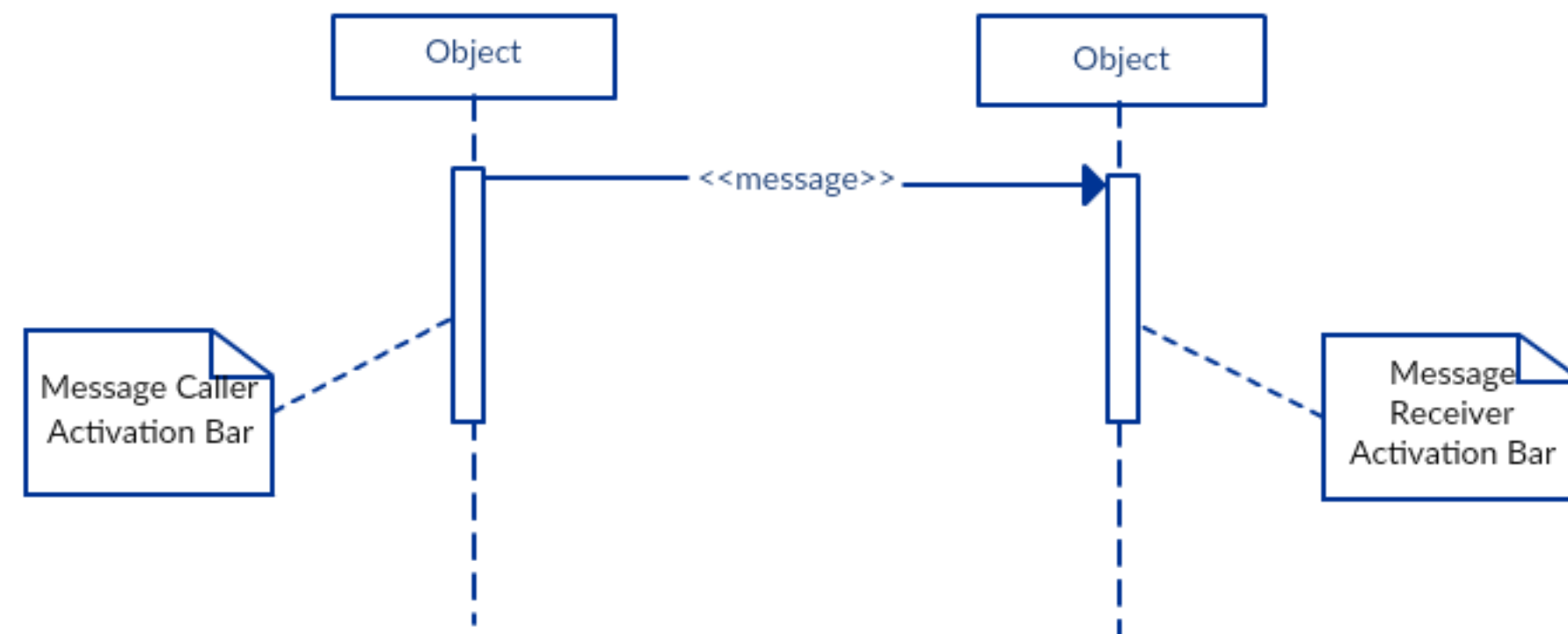
UML: Sequence Diagram - Control



A **Control** is a stereotyped Object that models a controlling entity or manager.

A Control organizes and schedules other activities and elements, typically in Analysis (including Robustness), Sequence and Communication diagrams. It is the *controller* of the MVC (Model-View-Controller) Pattern.

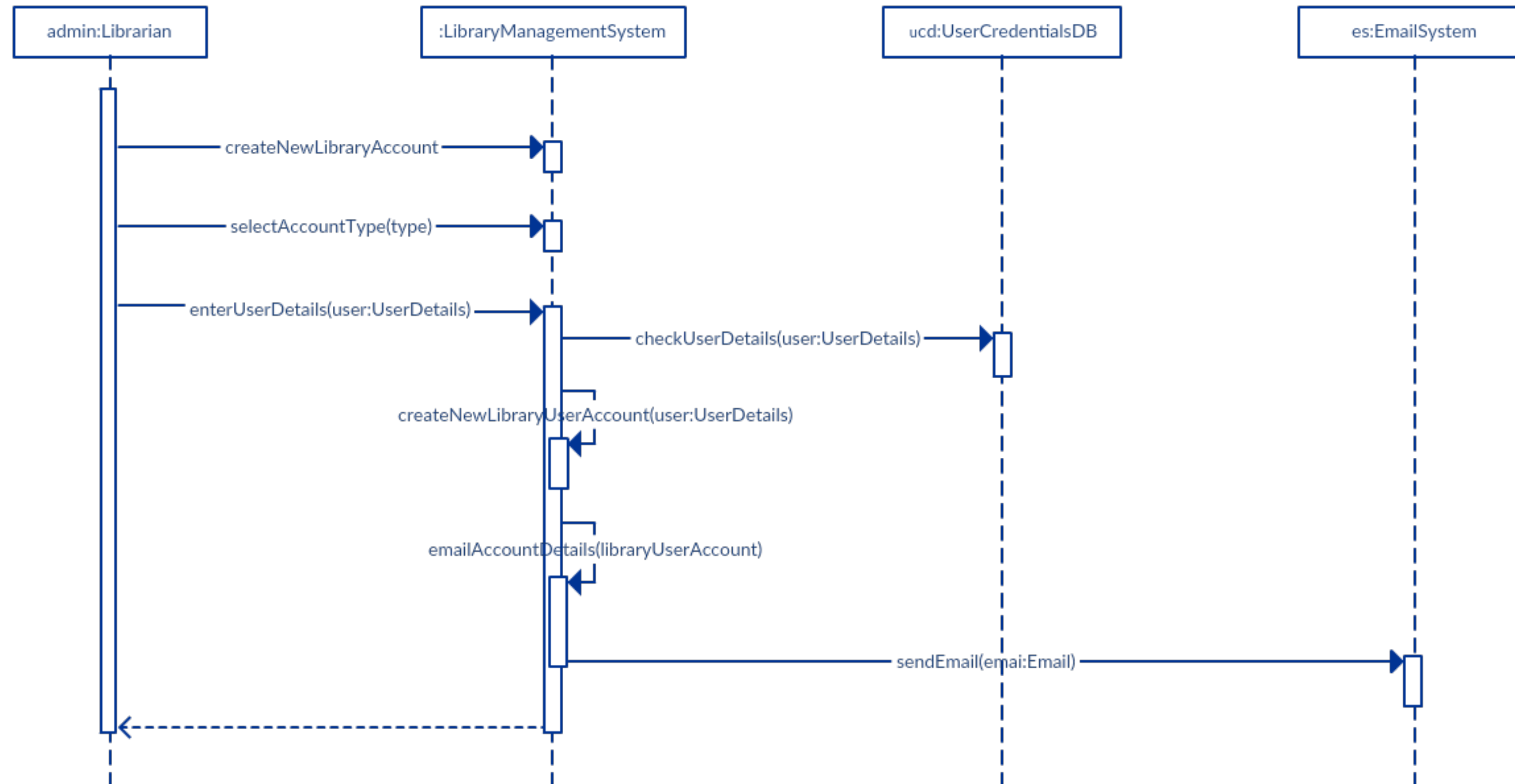
UML Sequence Diagram - Activation Bars



An **activation bar** is the box placed on the lifeline.

It is used to indicate that an object is active (or instantiated) during an interaction between two objects. The length of the rectangle indicates the duration of the objects staying active.

UML Sequence Diagram - Activation Bars



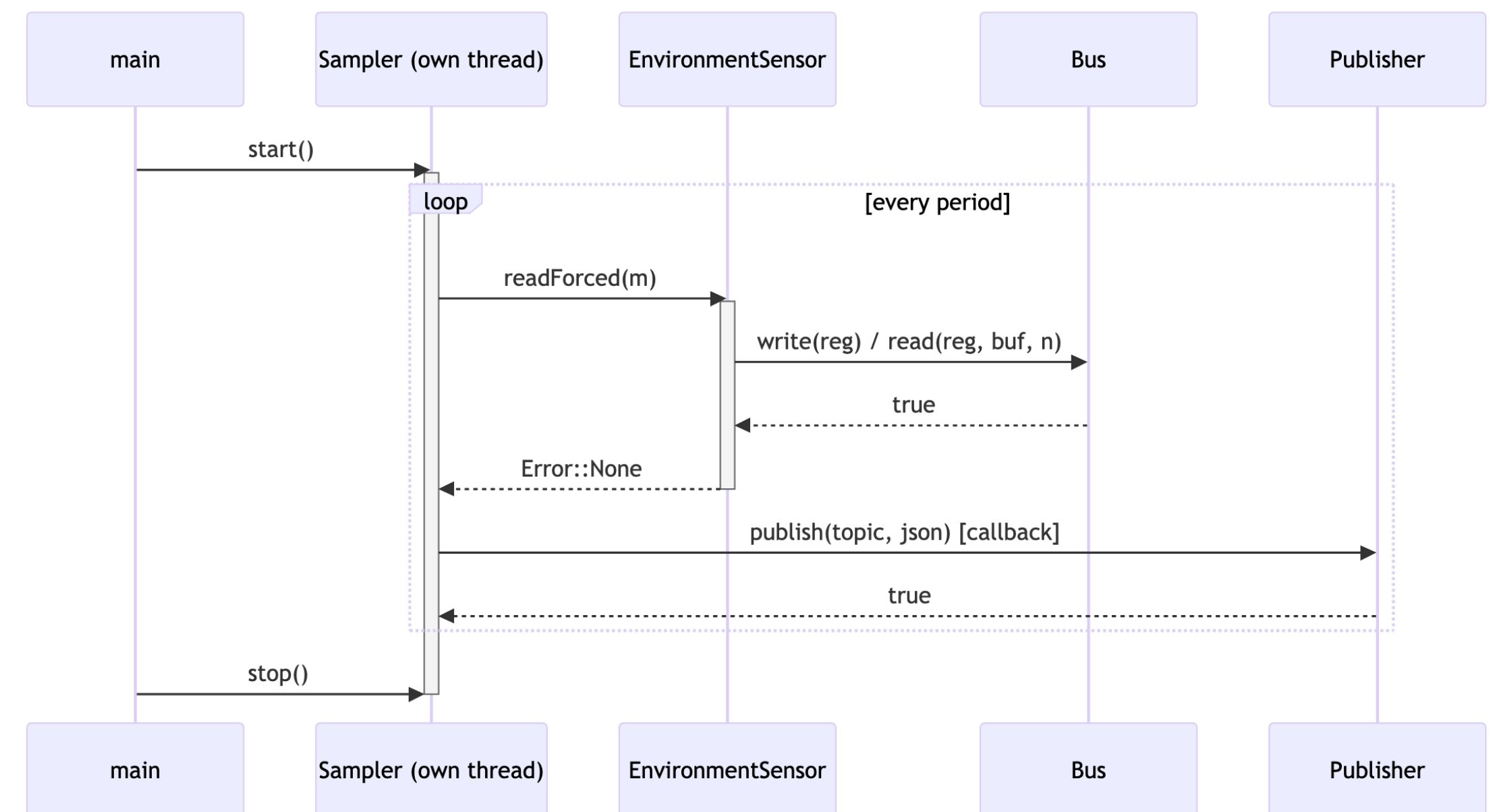
Sequence diagram: one sample-and-publish cycle

The activation bar on Sampler is the thread.

main appears twice: start() and stop().
Nothing in between.

The callback is the arrow from Sampler to Publisher.

The hop into the Bosch C code and back into Bus is inside readForced(): draw it when you design init() next week.



Live demo: broker down, broker up

Terminal 1 (Pi):

```
sudo apt install mosquitto mosquitto-clients libmosquitto-dev  
mosquitto_sub -h localhost -t 'han/ese/#' -v
```

Terminal 2 (Pi):

```
./build/station_main
```

Then:

```
sudo systemctl stop mosquitto      -> "publish failed (broker down?)", sampling continues  
sudo systemctl start mosquitto     -> messages resume (mosquitto_reconnect_delay_set)
```

Payload: {"t":21.37,"p":101325} topic: han/ese/<your-name>/bme280/state

Assignment 5: sampling thread, callback, MQTT (deadline Thu 8 Oct, tag v0.5-mqtt)

1. Sampler: own thread, period, callback, start / stop / destructor stops.
 2. Publisher interface with ConsolePublisher and MqttPublisher (libmosquitto, RAI).
 3. Composition root in main(): the lambda is the callback. No #ifdef, no globals, no singletons.
 4. EnvironmentSensor implemented by Bme280; FakeSensor runs the chain on your laptop.
 5. Topic han/ese/<name>/bme280/state, JSON payload, verified with mosquitto_sub.
 6. README: the threading paragraph (shared state, how it is protected).
 7. Sequence diagram of one cycle with the sampler thread's activation bar.
- Full hand-out, starter code and tags: github.com/AEAEmbedded/ESE_PROG -> Prog5/Assignments

Links on UML

Recommended resource: <https://creately.com/blog/diagrams/sequence-diagram-tutorial/>

https://www.tutorialspoint.com/uml/uml_interaction_diagram.htm

https://sparxsystems.com/enterprise_architect_user_guide/14.0/model_domains/otherelements2.html

Links on C++

<https://www.bfilipek.com/2019/12/threading-loopers-cpp17.html>

<https://www.geeksforgeeks.org/multithreading-in-cpp/>

<https://www.perforce.com/blog/qac/multithreading-parallel-programming-c-cpp>

<https://www.softwaretestinghelp.com/multithreading-in-cpp/>

C/C++ (< 11)

https://www.tutorialspoint.com/cplusplus/cpp_multithreading.htm

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Any questions?